



CANDIDATE
NAME

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CENTRE
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CANDIDATE
NUMBER

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0610/41

May/June 2023

1 hour 15 minutes

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

- 1 (a) Fig. 1.1 is a diagram of the digestive system.

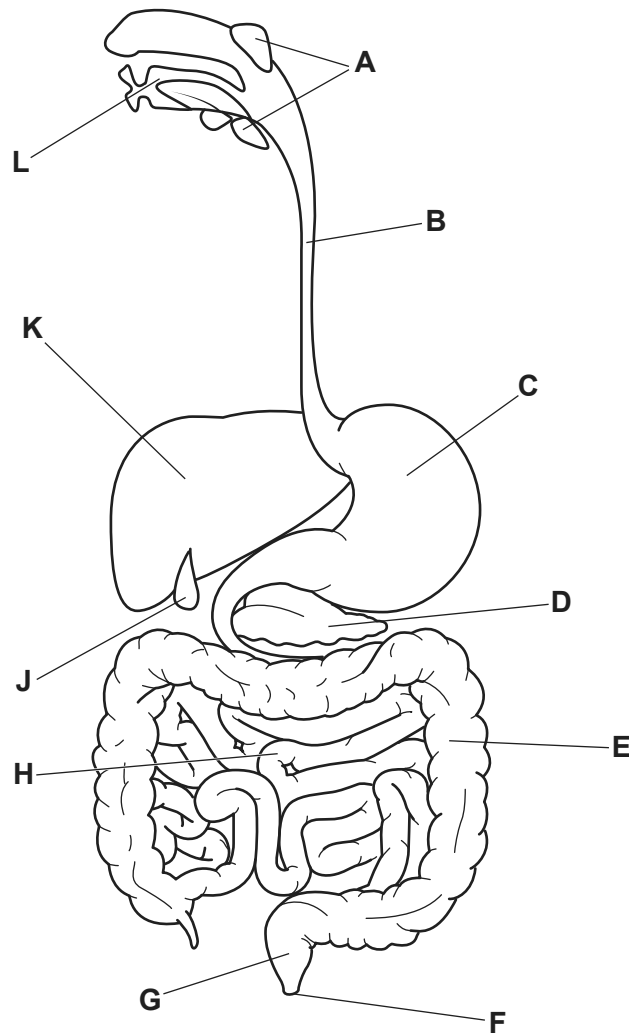


Fig. 1.1

Each letter may be used once, more than once or not at all.

State the letter of the part shown in Fig. 1.1:

that produces bile

that produces gastric juice

that produces urea

where maltose is digested

where trypsin acts.

[5]

(b) A student investigated the effect of bile on the digestion of fat in milk.

They set up three different test-tubes:

- test-tube **A** contained milk and bile
- test-tube **B** contained milk and lipase
- test-tube **C** contained milk, lipase and bile.

They used an indicator that is pink in alkaline solutions and colourless in acidic solutions. They added the same volume of indicator to each test-tube.

The student observed and recorded the colour of the contents of each test-tube at 0 minutes, 20 minutes and 40 minutes.

Table 1.1 shows the results of the investigation.

Table 1.1

test-tube	indicator colour observed		
	0 minutes	20 minutes	40 minutes
A	pink	pink	pink
B	pink	pink	colourless
C	pink	colourless	colourless

(i) Explain the results for test-tubes **B** and **C** in Table 1.1.

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.....

.....

.....

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..... [4]

(ii) Explain the purpose of test-tube **A** in Table 1.1.

.....

.....

.....

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..... [2]

(c) The action of lipase is affected by temperature.

Fig. 1.2 shows the axes for a graph of the effect of temperature on the activity of lipase.

Complete the graph by:

- drawing a line to show the expected effect of temperature on the activity of lipase
- adding a label line and a label to show the point at which all the lipase has been denatured.

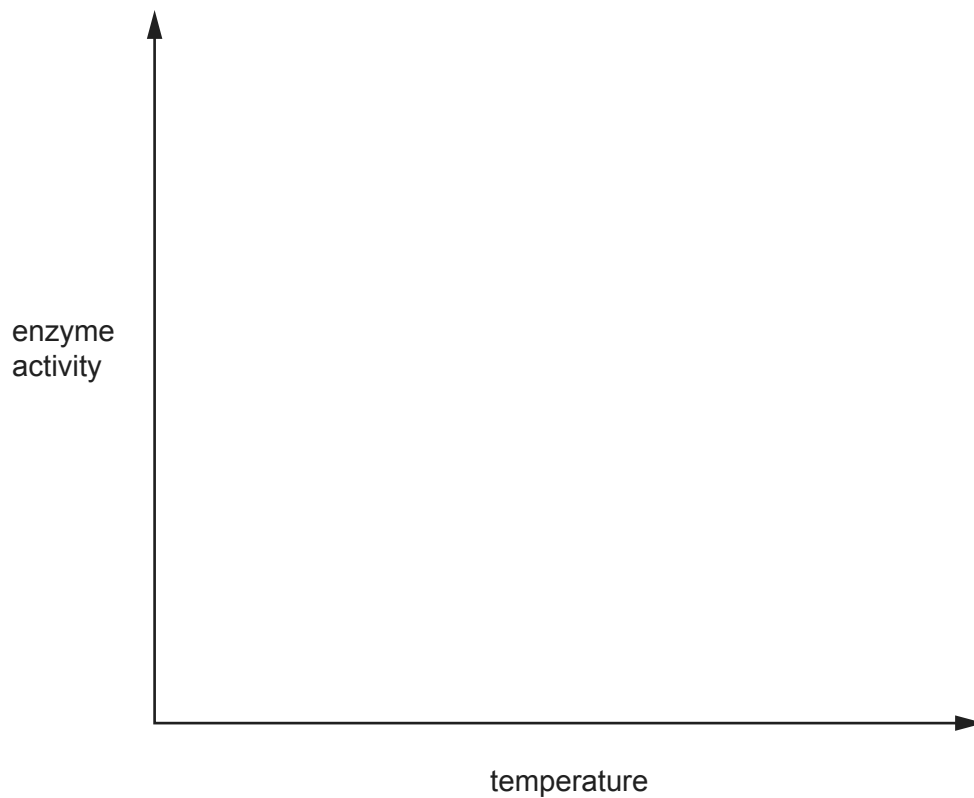


Fig. 1.2

[2]

(d) Explain why lipase cannot be used to catalyse the breakdown of proteins.

.....

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.....

.....

..... [3]

[Total: 16]

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- 2 (a) A student investigated osmosis in potato plant cells.

He immersed cubes of potato tissue in water and different concentrations of sucrose solution for 30 minutes.

The masses of the potato cubes were measured before and after immersion.

The percentage changes in mass were calculated.

Table 2.1 shows the results.

Table 2.1

concentration of sucrose solution / mol dm ⁻³	mass of potato cube before immersion / g	mass of potato cube after immersion / g	percentage change in mass
0.00	1.32	1.50	13.64
0.20	1.34	1.49	11.19
0.40	1.30	1.34	3.08
0.60	1.33	1.29	-3.01
0.80	1.22	1.12	-8.20
1.00	1.28	1.11	

- (i) Using the information in Table 2.1, calculate the percentage change in mass at 1.00 mol dm⁻³.

Give your answer to **two** decimal places.

Space for working.

..... %
[3]

- (ii) Using the information in Table 2.1, explain the difference in the results between the 0.6 mol dm^{-3} and the 0.8 mol dm^{-3} sucrose solutions.

Use the term water potential in your answer.

..... [5]

- (iii) Describe the expected appearance of a cell from a potato cube that has been immersed in distilled water for 30 minutes.

..... [2]

- (b)** Describe how the process of active transport differs from the process of osmosis.

..... [3]

- (c) State the type of plant cells that use active transport to absorb mineral ions from the environment.

..... [1]

- (d) Explain the effect of a lack of magnesium ions on the colour of plant leaves.

.....
.....
.....
.....
..... [2]

[Total: 16]

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- 3 (a) Fig. 3.1 is a photomicrograph of some cells lining the trachea.

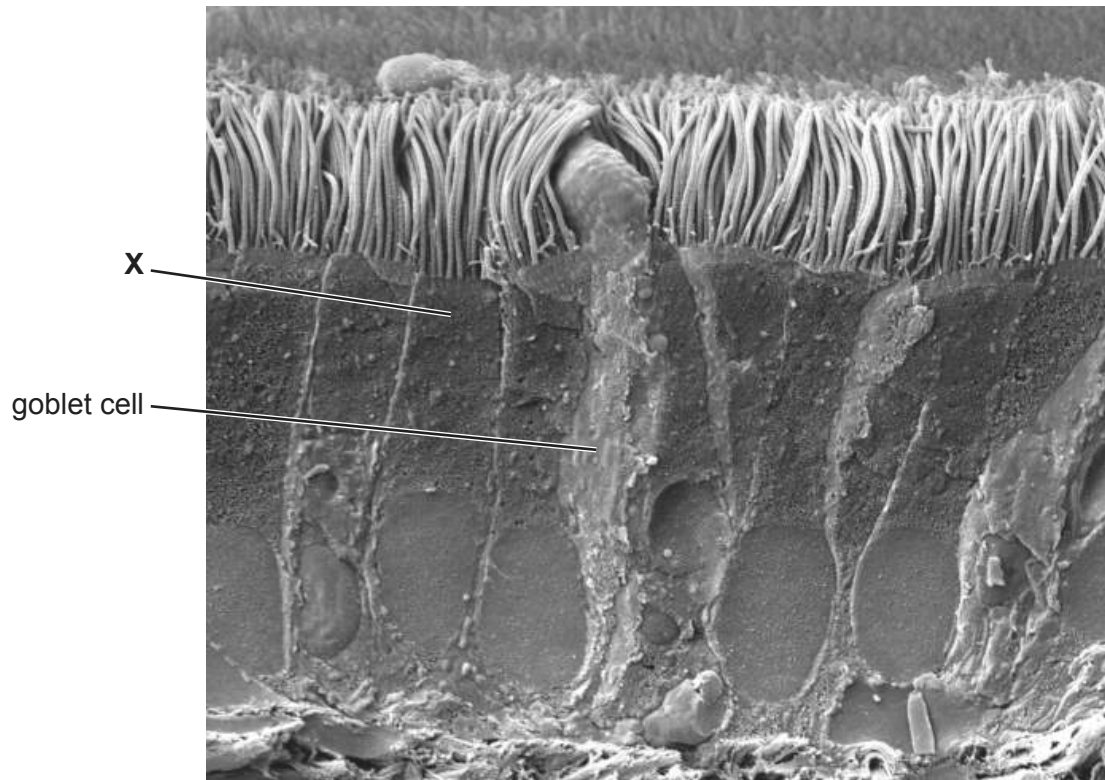


Fig. 3.1

- (i) Describe the role of goblet cells.

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.....

.....

..... [2]

- (ii) Explain how the cell labelled **X** in Fig. 3.1 is adapted for its function.

.....

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.....

.....

..... [2]

- (iii) State the name of **one other** part of the body where the type of cell labelled **X** in Fig. 3.1 is found.

..... [1]

- (b) Table 3.1 contains some features of the breathing system.

Complete Table 3.1 to show the actions of each feature of the breathing system that occur to cause inspiration.

Table 3.1

feature of the breathing system	action that causes inspiration
diaphragm	
external intercostal muscles	
pressure in the thorax	
ribs	
volume of the thorax	

[5]

- (c) State the name of the gas that is excreted by the breathing system.

..... [1]

- (d) Good ventilation is one feature of gas exchange surfaces.

State **two other** features.

1

2

[2]

- (e) State the name of the gas exchange surface in humans.

..... [1]

[Total: 14]

- 4 (a)** Fig. 4.1 shows the effect of light intensity on the rate of photosynthesis at different temperatures and concentrations of carbon dioxide.

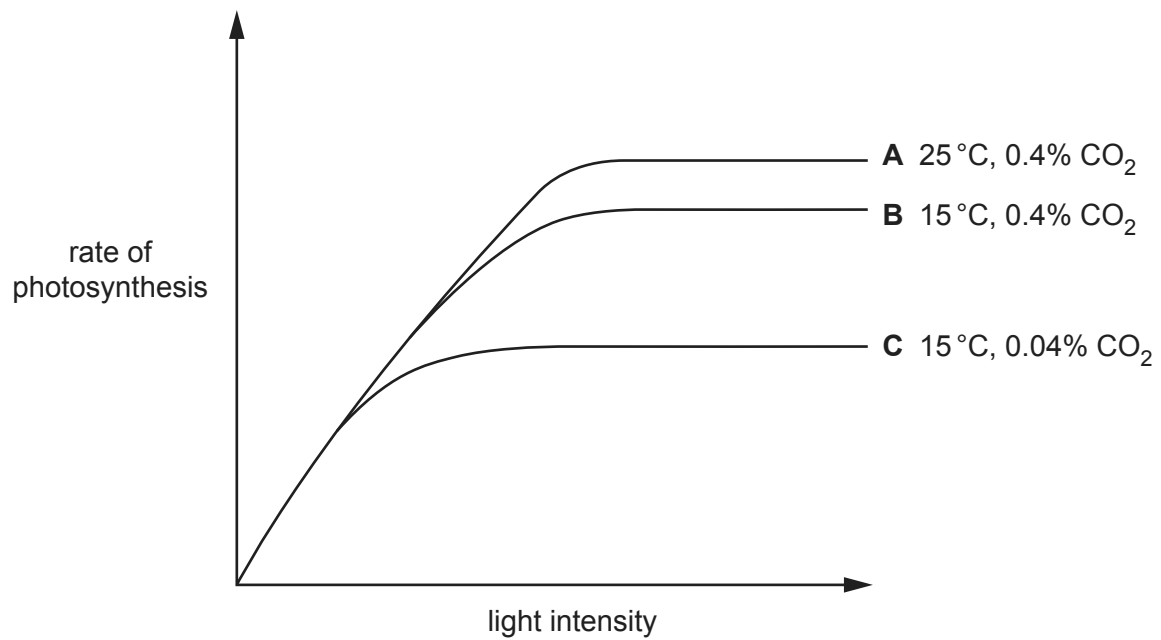


Fig. 4.1

Describe **and** explain the reasons for the shape of lines **B** and **C** in Fig. 4.1.

..... [6]

- (b) $C_6H_{12}O_6$ is one of the products of photosynthesis.

State the chemical formula of the **other** product.

..... [1]

- (c) Outline how the carbohydrates made during photosynthesis are used in plants.

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..... [4]

[Total: 11]

- 5 (a) Fig. 5.1 shows the stages involved in protein synthesis.

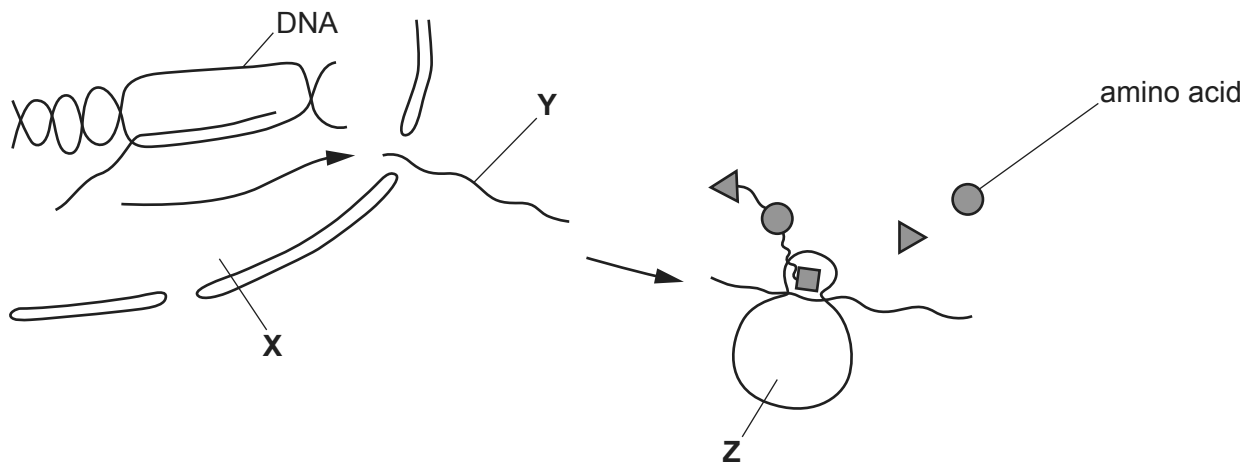


Fig. 5.1

- (i) State the names of the parts labelled **X**, **Y** and **Z** in Fig. 5.1.

X

Y

Z

[3]

- (ii) State what determines the sequence of the amino acids in the protein that is produced.

.....

.....

..... [1]

- (iii) Explain why the sequence of amino acids is important in the production of receptor molecules for neurotransmitters.

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..... [2]

- (b) Explain why body cells can have different specialised functions even though they contain the same genes.

.....

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.....

..... [2]

- (c) Allele frequency in a population can be changed by natural selection and artificial selection.

State the meaning of the term allele.

.....

..... [1]

- (d) Describe how artificial selection differs from natural selection.

.....

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..... [3]

- (e) Mutation causes formation of new alleles which increases genetic variation.

State **two other** sources of genetic variation in populations.

1

2 [2]

[Total: 14]

- 6 (a) A scientist monitored the changes in the pH in muscles before, during and after two minutes of vigorous exercise.

The changes in pH are caused by the production of lactic acid.

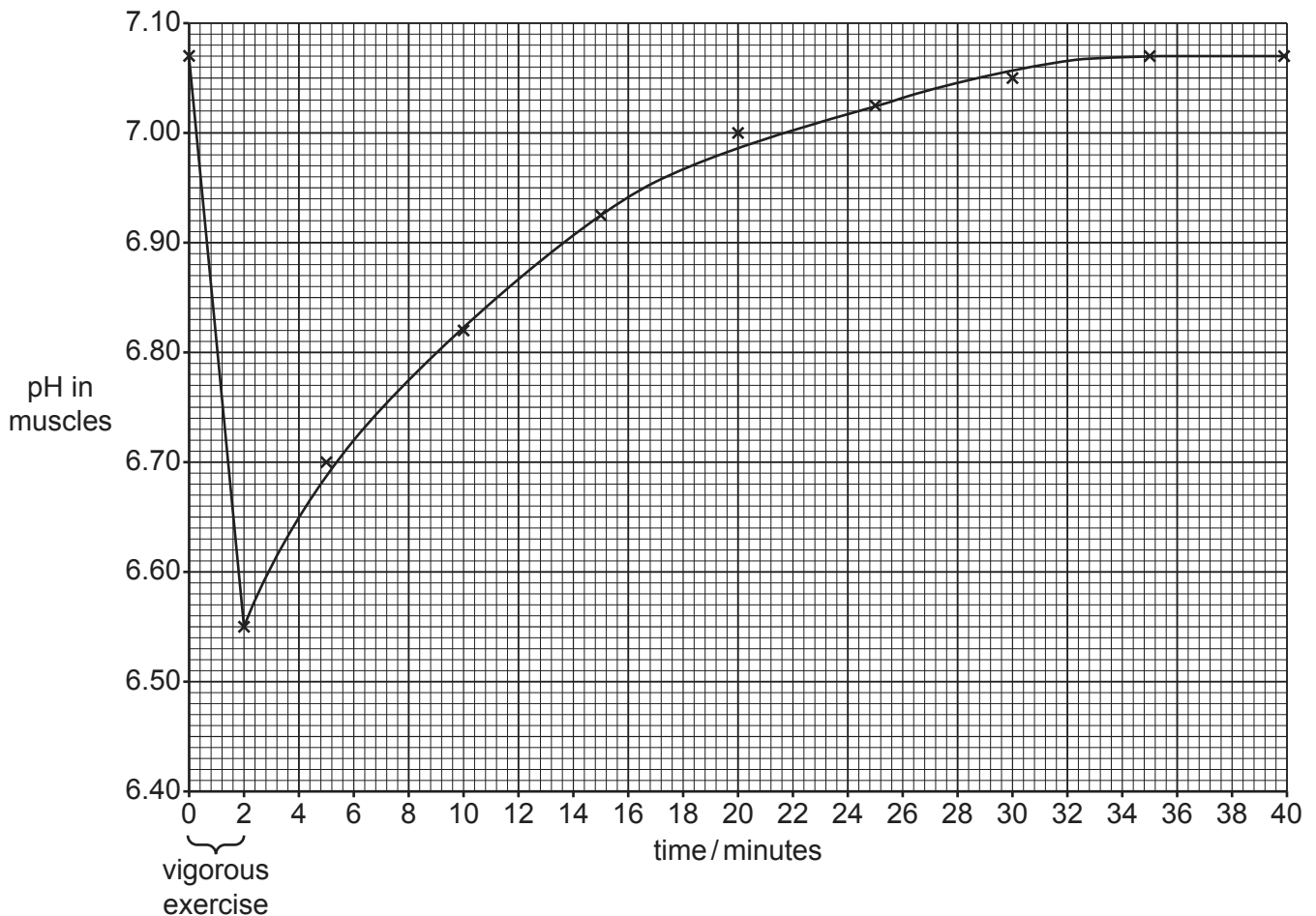


Fig. 6.1

Complete the sentences to describe **and** explain the results in Fig. 6.1.

The pH decreases from to during vigorous exercise.

There is not enough supplied to the muscles.

The body respire anaerobically. The lactic acid produced builds up in the muscles causing an debt.

It takes minutes for the muscle pH to return to its initial level after exercise.

The pH value increases after vigorous exercise has ended, as lactic acid is transported in the to the

During this time the breathing rate and rate remain high.

[6]

(b) Yeast can respire anaerobically.

(i) Complete the balanced chemical equation for anaerobic respiration in yeast.

$C_6H_{12}O_6$ [2]

(ii) Yeast belongs to the kingdom fungus.

State **one** cell component that is present in yeast cells but is absent in animal cells.

..... [1]

[Total: 9]

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